MAHARSHI OZA

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PROFESSIONAL SUMMARY

Computer vision and perception engineer with 2 years at Magna Electronics on surround-view ADAS perception. Hands-on with planar homography, camera calibration, bird's-eye-view generation, monocular 3D estimation, and KPI verification and validation in C++ and Python, plus DNN dataset preparation and evaluation. Background in hardware–software codesign and computer architecture.

CORE COMPETENCIES

Computer Vision: Planar Homography, Camera Calibration (Intrinsic/Extrinsic Parameters), Fisheye Undistortion, Bird's-Eye-View Generation, Feature Matching, Epipolar Geometry (8-Point Algorithm), RANSAC, Triangulation, 3D Reconstruction
ADAS Perception: Lane-Marker Detection / LDW, Front Object Detection, Trailer Hitch Detection, Near-Field Collision Detection, Surround-View Cameras, KPI Verification & Validation, Ground-Truth Comparison
Deep Learning: Data Annotation, Training/Test Dataset Preparation & Validation, PyTorch, DNN vs Classical Pipeline Evaluation
Languages & Tools: C++, C, Python, OpenCV, SystemVerilog, Verilog, SystemC, GDB, MISRA C/C++ Compliance, Code/Branch/Line Coverage, Git, Linux
Computer Architecture: RISC-V, Pipelining, Out-of-Order Execution, Memory Hierarchy, Hardware–Software Codesign, Hardware Accelerators

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Dec 2023 – Aug 2024**

- Built an out-of-order AXI4 scoreboard in SystemVerilog/UVM (over 90% functional coverage) and contributed to a multi-port packet router RTL design.

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Dec 2023**

- Worked across surround-view ADAS perception modules: lane-marker detection (LMD, later lane departure warning, LDW), front object detection, trailer hitch detection, and near-field collision detection (NFCD).
- Developed verification and validation code for perception KPIs, including lane-marker and lane-type classification, lane width and inter-lane distance, comparing algorithm outputs against annotated ground truth.
- Implemented bird's-eye-view (BEV) generation in C++ from the vehicle's four surround-view cameras using planar homography with camera intrinsic and extrinsic parameters, built on Magna's in-house vision libraries.
- Contributed to migrating the perception code base from 32-bit to 64-bit, resolving data-type, portability and build issues across modules.
- Supported the move from classical image processing to DNN-based perception: annotated images, prepared training datasets, validated test datasets, and compared DNN results against the traditional image-processing pipelines.
- Implemented code, branch, and line coverage for perception modules and the KPI framework to expose validation gaps; root-caused edge-case failures with GDB; all development followed MISRA C/C++ compliance.

PROJECTS

Trailer Coupler Height Estimation Using a Monocular Fisheye Camera | Magna Electronics | C++, Python, OpenCV

- Estimated the 3D height of a trailer coupler from a single rear-view fisheye camera to support automated trailer hitching, using structure-from-motion across frames as the vehicle moved.
- Undistorted fisheye images to a pinhole model using camera intrinsics, matched features across frames with OpenCV, estimated camera geometry with the 8-point algorithm inside RANSAC to reject outliers, and triangulated matches to reconstruct 3D points and recover coupler height.

Decoder-Only Transformer Language Model from Scratch | Python, PyTorch, CUDA

- Built a fully functional decoder-only transformer (causal multi-head attention, byte-level BPE tokenizer, training and generation pipeline) in PyTorch and trained it on TinyStories on an RTX 2080 Ti; ran controlled experiments where scaling from ~5M to ~13M parameters cut validation perplexity from 38.2 to 22.3, and post-norm reduced it to 34.2 vs. pre-norm.

Stride Value Predictor for RISC-V OOO Pipeline Simulator | C++

- Developed a cycle-accurate RISC-V out-of-order simulator and integrated a Stride Value Predictor with VPQ/checkpoint recovery; added a Global Stride History Table with LRU replacement, improving harmonic-mean IPC from 1.91 to 2.00 across 15 benchmarks within a 27 KB storage budget.

Cache and Memory Hierarchy Simulator | C++

- Developed a configurable trace-driven L1/L2 cache simulator in C++ with parameterized block size, capacity and associativity, LRU replacement, and a stream prefetcher, reporting miss rates and memory traffic across cache configurations.

FIR Filter Accelerator via High-Level Synthesis | SystemC, Catapult HLS, RISC-V Rocket-Chip

- Implemented a 16-tap FIR filter accelerator in SystemC and synthesized it with Catapult HLS at a 5 ns clock, exploring architectures that reduced latency from ~3.5K to 69 cycles.

Floating-Point Matrix Multiplier | SystemVerilog, Design Compiler

- Designed an SRAM-based single-precision (32-bit) floating-point matrix multiplier around a DesignWare floating-point MAC with FSM-driven address generation and write-back; passed 256/256 result checks across 4 test cases and closed timing at 10 ns in Design Compiler (Nangate 45 nm, ~12.1K cells, ~13.7K μm^2).

UVM Verification of a Set-Associative Cache Controller | SystemVerilog, UVM, SVA, Python

- Built a UVM environment with CPU, flush and reactive-memory agents, a reference-model scoreboard and SVA checks; Python regression flow merging coverage, with 0 data mismatches across 1,000+ accesses under injected errors.

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: Hardware–Software Codesign, Advanced Microprocessor Architecture, ASIC and FPGA Design with Verilog, Advanced Verification using UVM

B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**
Relevant Coursework: Digital Image Processing, Digital Signal Processing, Signals and Systems, Object-Oriented Programming with C++, Microprocessors